Zener Diode as Voltage Regulation

Aim:

To simulate the line and load Regulation operation using zenar diode.

Apparatus Required:

Laptop with proteus Software.

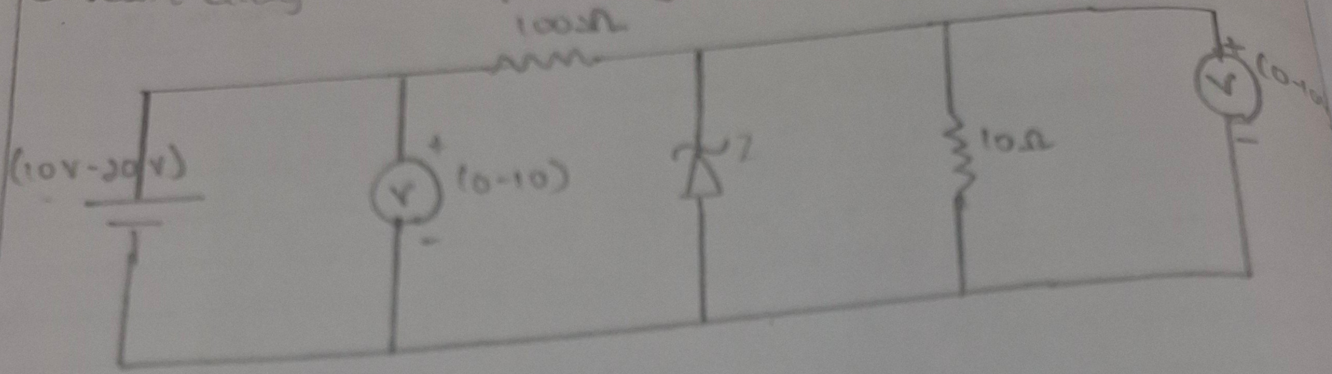
Theory:

A heavily dopped pn Junction diode is called zener diode. Due to heavily dopped nature zener diode works under forward bias and reverse bias condition.

Zener diode under reverse bias condition is used as a voltage regulation. Even if The input voltage changes The output is Constant. This is called line regulation.

If The Load resistence changes The output voltage is Constant this is called load regulation.

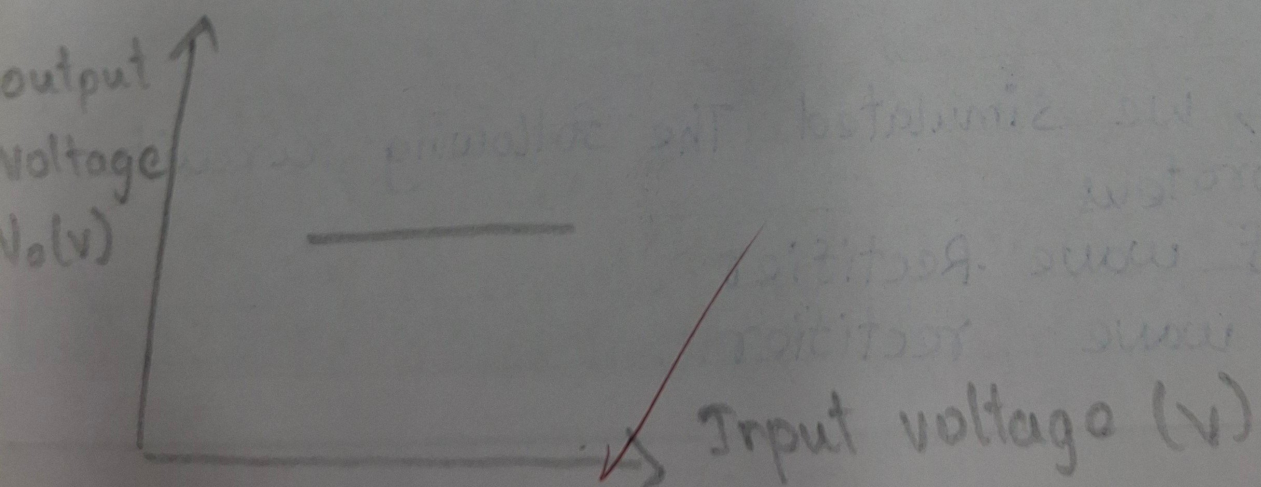
Circuit diagram: Line regulation.



Tabulation

S.No	Input voltage	output voltage V_o (V)
1	+ 12.0	+ 5.14
2	+ 13.0	+ 5.16
3	+ 14.0	+ 5.17
4	+ 15.0	+ 5.17
5	+ 16.0	+ 5.18
6	+ 17.0	+ 5.19

Model Graph:



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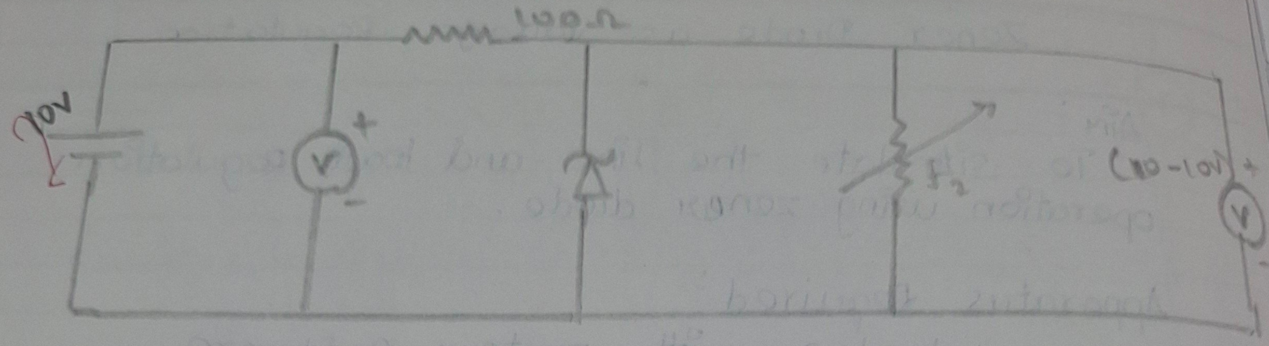
procedure :

1. Drag The required components from The proteus Library.
2. connect The components (required) as per circuit diagram,
3. For The ~~r~~ Line regulation vary The input voltage and not The output voltages in the tabulation .-Draw the graph between input and output voltages
4. For Load regulation vary the Load regulation vary the Load resistence and note The output voltage in tabulations. Draw The graph between Load resistence and outputs voltage.

Circuit Diagram:

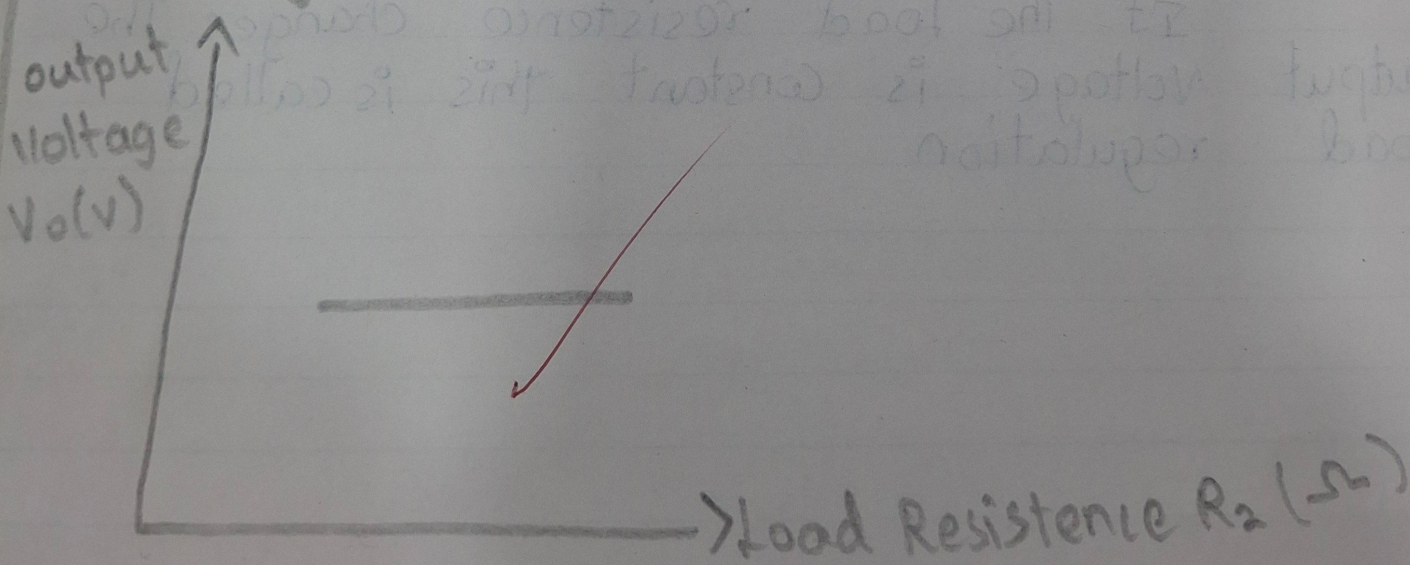
Load Regulation



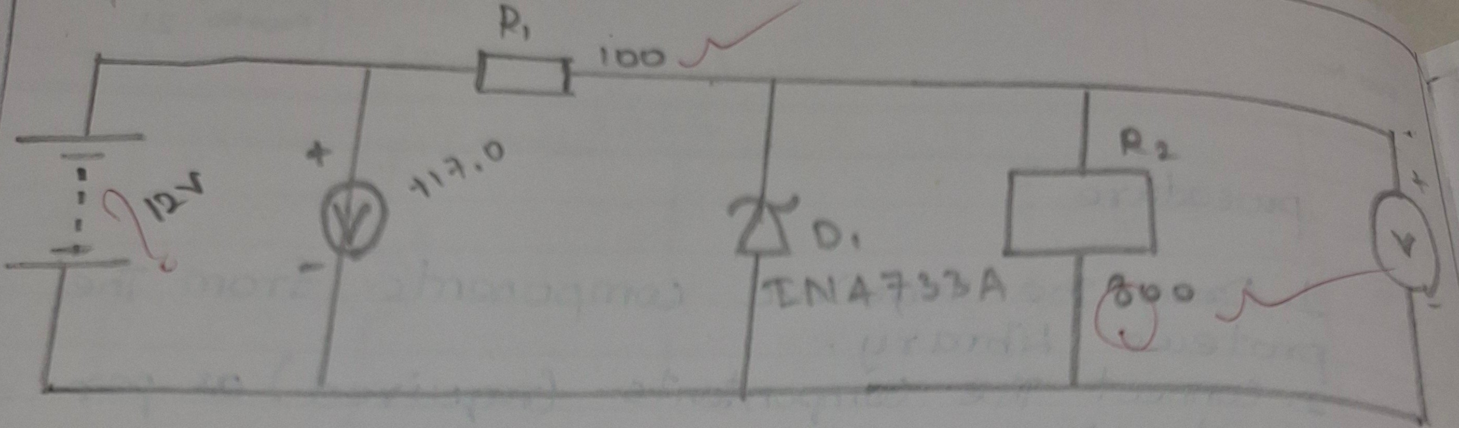
Tabulation:

S.NO	Load Resistence R_2 (Ω)	output voltage V_o (V)
1	100 Ω	5.19V
2	200 Ω	5.20V
3	300 Ω	5.20 V
4	400 Ω	5.20 V
5	500 Ω	5.20 V
6	600 Ω	5.21 V

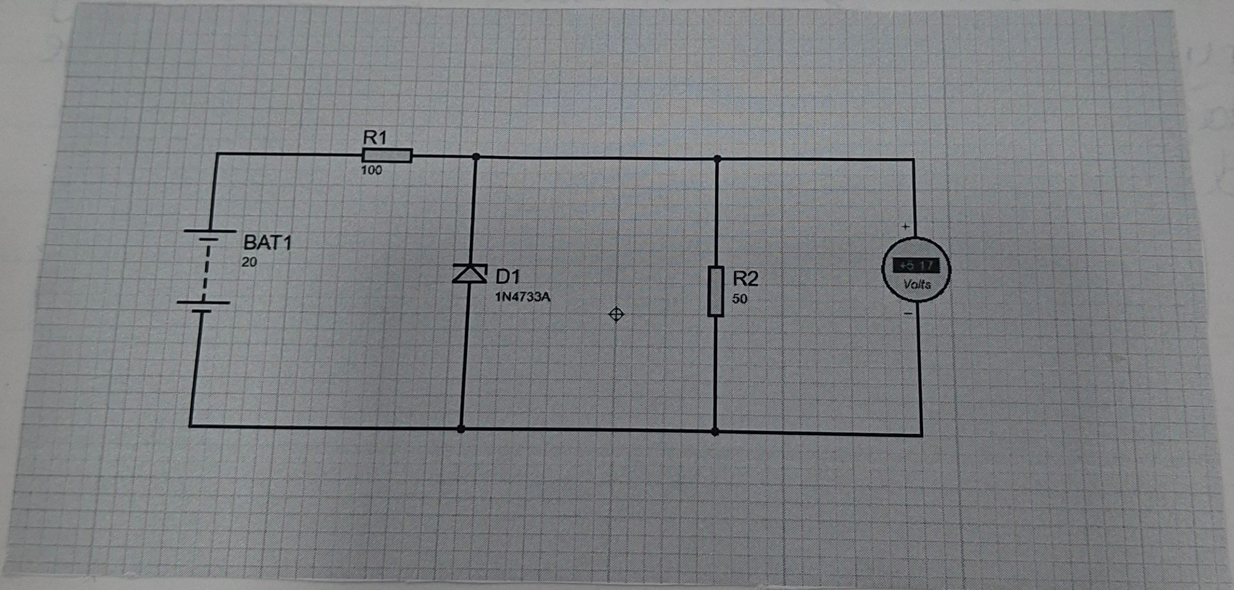
Model diagram:



output:



R1
100
+
12V
17.0
D1
IN4733A
R2
500



R1
100
BAT1
20
D1
1N4733A
R2
50
+5.17
Volts

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Result;

Thus simulation of The Line and Load regulation operation of zener diode is successfully completed.